Karthik Uppuluri Electrical and Computer Engineering

ELEC 547 F13 Assignment # 3

Acknowledgement: I have read the reading part of the assignment

Part A

Question: 1

Given equations

$$x = alpha_x \times \left(\frac{X}{Z}\right) + S \times \left(\frac{Y}{Z}\right) + P_x$$

$$y = alpha_y \times \left(\frac{Y}{Z}\right) + P_y$$

The intrinsic parameter matrix is given by

$$K = \begin{bmatrix} 3322.8 & -384.9 & 768 \\ 0 & 2886.9 & 512 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow alpha_x = 3322.8;\ alpha_y = 2886.9$$

From [(1)] $S = alpha_c \times alpha_x$ where alpha_c is skew coefficient, so the skew coefficient in our case is -0.1158, so the angle in degrees would be -6.6^0. Code for part A is attached in the file

Part B

1. 3D to 2D point correspondences
   Six 2D points have been chosen using ginput function for which 3D point correspondences are known. Each column corresponds to a point.

   2D matrix:

   $$\begin{bmatrix} 1121.5 & 593.5 & 589.5 & 1113.5 & 1089.5 & 939.5 \\ 591.5 & 847.5 & 127.5 & 145.5 & 847.5 & 429.5 \end{bmatrix}$$

   3D matrix:

   $$\begin{bmatrix} 3 & 0 & 0 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 & 0 & 1 \\ 0 & -3 & 0 & 0 & -3 & 0 \end{bmatrix}$$

2. Projection matrix, K, Rotation matrix and translation matrix

$$P = \begin{bmatrix} 0.2208 & 0.0318 & -0.0493 & 0.6513 \\ 0.0028 & -0.1748 & -0.1523 & 0.6855 \\ 0.0000 & 0.0000 & -0.0001 & 0.0012 \end{bmatrix}$$

$$K = \begin{bmatrix} 3156.4 & -16.7 & 842.0 \\ 0 & 3242.8 & 689.3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} -0.9999 & -0.0097 & -0.0073 \\ -0.0120 & 0.8741 & 0.4857 \\ -0.0017 & -0.4857 & 0.8741 \end{bmatrix}$$

$$T = \begin{bmatrix} 1.5066 \\ 0.5267 \\ -16.6973 \end{bmatrix}$$

Code for part B is attached in the zip file.

3. A. Direct Linear Transform: DLT is a method of solving similarity equations as a set of linear homogenous equations [2].

Problem statement: Solve for 'A' such that $\alpha x = AY$ (equation 1), in this case α is an unknown scaling factor, x is a matrix (each column is an independent point) of points on the image plane and Y is a matrix (each column is an independent point) of points in the scene.

Algorithm:
The algorithm is repeated for each column in x and Y, i.e. for each point correspondences.

Homogenous Point on Image Plane $x_k = \begin{pmatrix} x_{1k} \\ x_{2k} \\ 1 \end{pmatrix}$, Homogenous Point in the scene $y_k = \begin{pmatrix} y_{1k} \\ y_{2k} \\ y_{3k} \\ 1 \end{pmatrix}$

K = 1, 2……N where N is the number of correspondences available.

Goal: Find $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$

a. To compensate for the unknown scalar factors multiply with $x_k^T H$ on both sides of equation (1), k denotes the column number and H is an anti-symmetric matrix.

b. Resultant equation is $x_k^T H A y_k = 0$

c. And the resultant matrix can be written as a set of homogenous equations, and can be represented as $b_k^T a = 0$.

d. From the above equation' a' lies in the null space of matrix B (all the columns included) and can be solved using singular value decomposition (SVD).

e. Once we obtain the solution 'a', since 'a' is row vector it should be reshaped into 3x4 matrix to obtain the camera projection matrix.

Considering the cross product of αx and Ay instead of multiplying with $x_k^T H$ on both sides gives the same result.

B. Minimum number of point correspondences required is 6 (exactly 5.5 i.e. 5 full correspondences and 1 one correspondence in at least x or y)
If we have one point correspondence (i.e. X1 -> x1) we have two linearly independent equations, one for x and other for y. Since we have 11 degrees of freedom (5 for calibration matrix, 3 for rotation and 3 for translation) we at least need 11 equations to compute those parameters, so we at least need 6 point correspondences so that we can form twelve equations.

C. How to obtain K R t from Projection matrix

$$\text{ProjectionMatrix } P = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = K[R\ t]$$

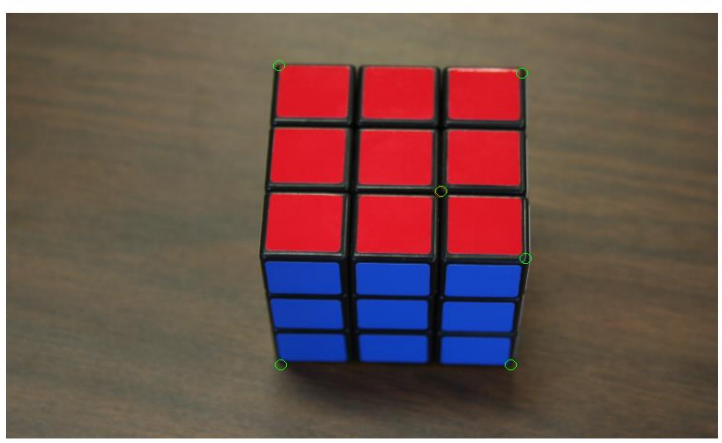
$$\text{Choose First } 3\times 3 \text{ sub-matrix}$$

$$M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = KR$$

$$RQ(M) \ \ (\text{using RQ decomposition}) = K\ \ R$$

$$t = M^{-1} P_4$$

4. Error calculation (Error = 0.0842 for the hard coded value)



Green circles indicate actual 2D points chosen using ginput function and red circles indicate re-projected points.

Part B

1. Camera calibration results
   Case.1 Window size [5 5]

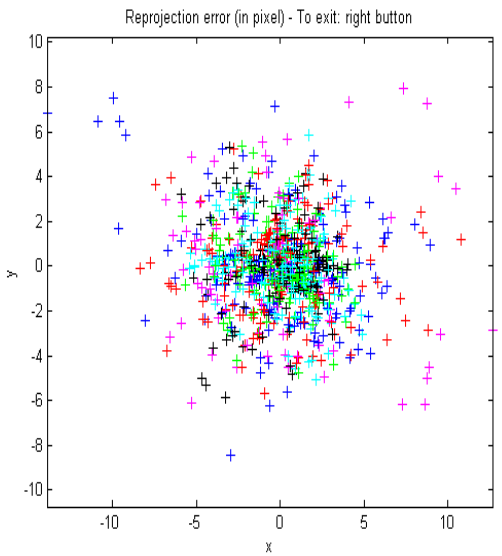


Figure.1 Re-projection error

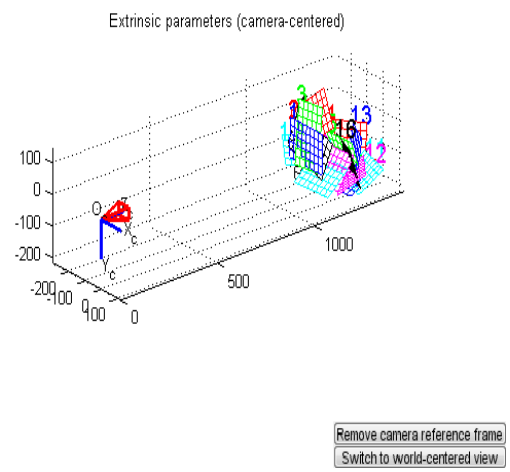


Figure.2 Extrinsic parameters

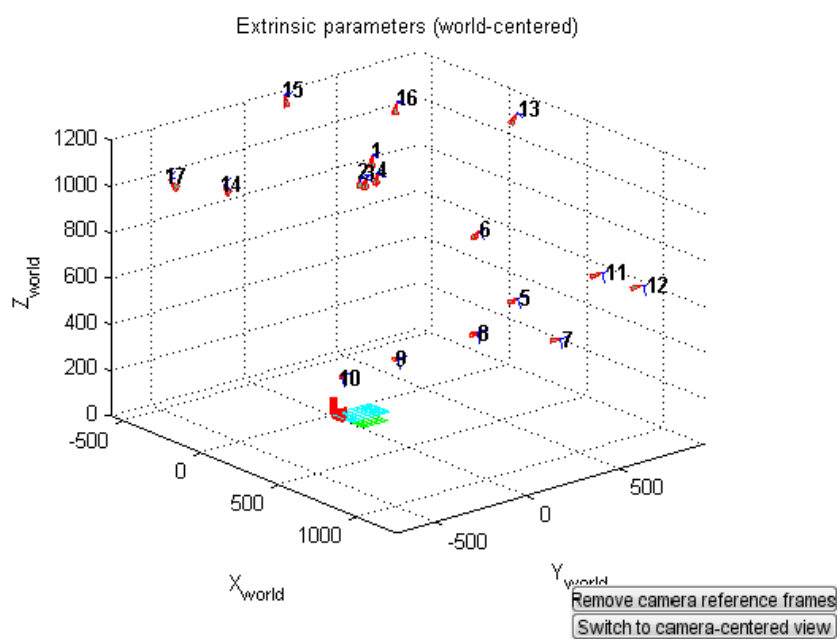


Figure.3 Extrinsic parameters world-centered view

Camera parameters (Window size [5 5])
Initialization of the intrinsic parameters - Number of images: 17

Calibration parameters after initialization:

Focal Length: fc = [5575.66495 5575.66495]
Principal point: cc = [1727.50000 1151.50000]
Skew: alpha_c = [0.00000] => angle of pixel = 90.00000 degrees
Distortion: kc = [0.00000 0.00000 0.00000 0.00000 0.00000]

Main calibration optimization procedure - Number of images: 17
Gradient descent iterations:
1...2...3...4...5...6...7...8...9...10...11...12...13...14...15...16...17...18...19...20...21...22...23...24...25...26...27...28...29...30...done
Estimation of uncertainties...done

Calibration results after optimization (with uncertainties):

Focal Length: fc = [7128.10158 7029.96465] ± [242.03219 231.51423]
Principal point: cc = [2151.31351 942.36105] ± [291.36106 191.85054]
Skew: alpha_c = [0.00000] ± [0.00000] => angle of pixel axes = 90.00000 ± 0.00000 degrees
Distortion: kc = [0.39661 -4.70031 -0.00503 0.02920 0.00000] ± [0.19824 3.08689 0.01360 0.02346 0.00000]
Pixel error: err = [2.92012 2.15023]

Note: The numerical errors are approximately three times the standard deviations (for reference).

Case.2 Window size [8 8]
Camera Parameters
Main calibration optimization procedure - Number of images: 17
Gradient descent iterations:
1...2...3...4...5...6...7...8...9...10...11...12...13...14...15...16...17...18...19...20...21...22...23...24...25...26...27...28...29...30...done
Estimation of uncertainties...done

Calibration results after optimization (with uncertainties):

Focal Length: fc = [7783.97776 7667.37152] ± [327.56298 311.11414]
Principal point: cc = [2047.81533 1042.75503] ± [392.84944 283.64030]

Skew: alpha_c = [0.00000] ± [0.00000] => angle of pixel axes = 90.00000 ± 0.00000 degrees

Distortion: kc = [-0.30511 5.74993 0.01566 0.02347 0.00000] ± [0.25742 6.34769 0.01130 0.01526 0.00000]

Pixel error: err = [2.80332 3.03687]

Note: The numerical errors are approximately three times the standard deviations (for reference).

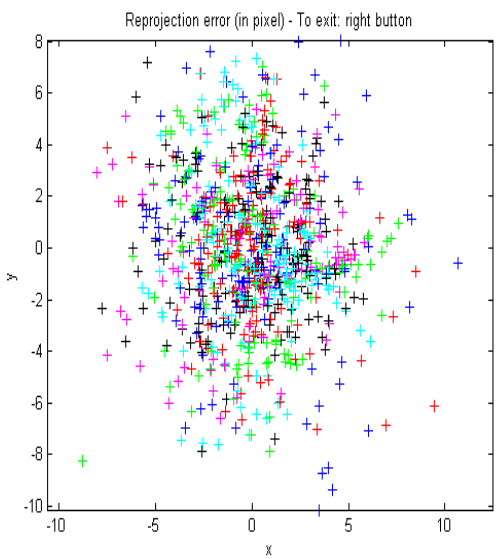


Figure.4 Re-projection error

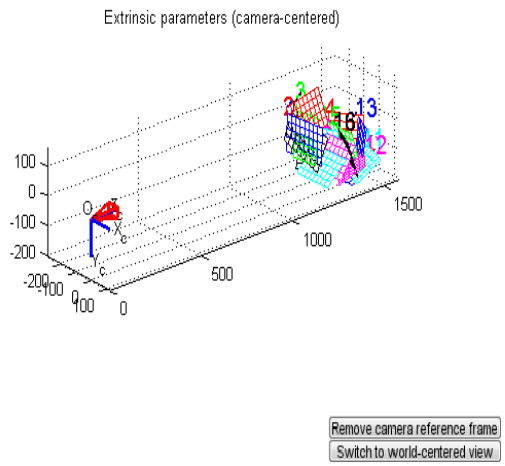


Figure.5 Extrinsic parameters

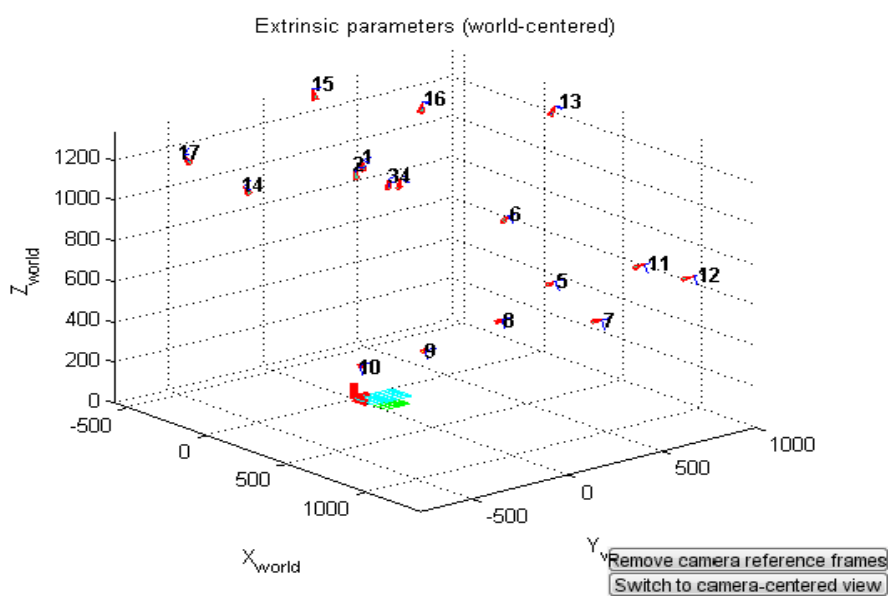


Figure.6 Extrinsic parameters world-centered view

Summary of "A flexible new technique for camera calibration"(Zhengyou Zhang)[3]

Zhang's algorithm finds feature points of each planar pattern, followed by homographies for each pattern and finally the camera parameters are recovered from homographies. Once the camera parameters are obtained gradient methods are used to optimize the results obtained by minimizing the difference between the actual and projected points.

Homography between the model plane and its image is calculated assuming that the model plane is at infinity. Two Constraints on the intrinsic parameters are obtained by using the property that the columns of the rotation matrix are orthonormal. Once we obtain the solution (intrinsic parameters using absolute conic matrix) for the two constraints we can recover all the camera parameters (intrinsic + extrinsic) from the solution. The number of images required to compute this solution depends on the number of camera parameters we already know. Once the camera parameters (intrinsic) are obtained motion of the camera (extrinsic parameters) can also be obtained by computing rotation and translation matrix. Finally, gradient methods are used to optimize the parameters obtained by minimizing the difference between the projected points and the actual points

References:

[1]. http://www.vision.caltech.edu/bouguetj/calib_doc/htmls/parameters.html

[2]. Appendix: DLT algorithm (direct Linear Transformation) From: Multiple View Geometry in Computer Vision, by R. Hartley and A. Zisserman, Academic Press, 2002

[3]. Z. Zhang. A flexible new technique for camera calibration. IEEE Transactions on Pattern Analysis and Machine Intelligence, 22(11):1330-1334, 2000.